

Barton Springs Salamanders

Jamhad Byers & Nick Gonzalez

2025-03-02

Contents

1	Barton Springs Salamanders	1
1.1	Original Variables	2
1.2	Created Variables	2
2	Methods	2
2.1	Loading Packages	2
2.2	Importing the Data	3
2.3	Choosing 4 Sites for Analysis	3
2.4	Understanding Filter Fraction	4
2.5	Choosing Parameters to Review	4
2.6	Associated Units List	5
3	Data Cleaning	6
4	Results	7
4.1	Summary Stats	7
4.2	Survey Area by Site	8
4.3	Samople Area Over Time	9
4.4	DO Over Time	11
4.5	DO by Site	13
4.6	Salamanders	14
4.7	Flowcodes	15
4.8	Inquiries Before Modeling	15
5	Modeling	20
6	Discussion	23
7	Reflection, Acknowledgements, & References.	24

1 Barton Springs Salamanders

The Barton Springs Salamanders DO and Flow data-set from the City of Austin provides measurements of dissolved oxygen (DO) and spring flow rates at Barton Springs, an important habitat for the endangered Barton Springs Salamander (*Eurycea sosorum*). The data-set includes time-series records of DO concentrations (in mg/L) and spring flow (in cubic feet per second) collected from various sites within the Barton Springs complex. These environmental factors are critical indicators of water quality and habitat health for the salamander population. By analyzing trends in DO levels and flow rates, researchers and conservationists can assess

the impact of environmental changes, groundwater conditions, and human activity on the ecosystem.

This dataset is interesting due to its relevance in the broader conversation about climate change and the methods used to keep track of the nuanced and widespread effects. For our purposes here, the data will be used to assess dissolved oxygen levels, salamander counts, flow severity, and sample areas of four sites in the Austin area. Each row represents a measurement on a particular day at a particular site.

1.1 Original Variables

`site_name` is, as the name suggests, the name of the site for that particular instance of data collection.

`sample_date` is also as its name suggests, in that it lists the date that sample was taken.

`parameter`, `result` & `unit` are the measurement, numeric value of that measurement, and the units of that measurement respectively.

1.2 Created Variables

`DO` is created from the `Dissolved Oxygen` value in `PARAMETER`, and contains the numeric values of the measurement.

`Flowcode` is created from the `FLOW SEVERITY CODE` (1=NONE;2=LOW;3=NORM;4=FLOOD;5=HIGH;6=DRY) value in `PARAMETER`, and is a categorical representation by the Texas Commission on Environmental Quality (TCEQ) of the type of flow observed. The numbers correspond to their listed type.

`salamanders` is created from the `NUMBER OF AUSTIN BLIND SALAMANDERS PHOTOGRAPHED` value in `'PARAMETER'`, and is a count of how many salamanders were seen during that particular measurement.

Finally, `area` is created from the `AREA SAMPLED` value in `PARAMETER`, and is a measure of the total square footage that was surveyed for a particular site on a particular day.

The data is provided by the Barton Springs Salamanders DO and Flow data-set from the City of Austin and is a “view” of the Water Quality Sampling Data dataset - in that the Parameters and sites and projects are limited to those related to this particular subject (Barton Springs Salamanders, Flow and DO).

Before cleaning, the set contains 50,068 rows and 22 columns.

2 Methods

This analysis investigates how dissolved oxygen (DO) levels vary across different Barton Springs sites, and whether that variation is confounded by differences in survey effort (area sampled) or salamander detection rates across sites. An initial question regarding whether filter fraction (dissolved vs. total) affects DO readings was set aside, as filter type was inconsistently recorded across site visits and would have required a separate data structure to address properly. This is a natural extension for future work.

2.1 Loading Packages

```
# Upload / install packages
library(tidyverse)
```

```
## Warning: package 'lubridate' was built under R version 4.4.1
```

```
library(plotROC)
```

2.2 Importing the Data

```
water <- read_csv("Barton_Springs_Salamanders_DO_and_Flow_20250305.csv")
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,  
## e.g.:  
##   dat <- vroom(...)  
##   problems(dat)
```

```
# data preview
```

```
head(water)
```

```
## # A tibble: 6 x 22  
##   WATERSHED    SAMPLE_DATE SITE_NAME LON_DD_WGS84 LAT_DD_WGS84 SITE_TYPE MEDIUM  
##   <chr>        <chr>      <chr>      <dbl>      <dbl> <chr>    <chr>  
## 1 Barton Creek 02/25/2025 ~ Barton S~    -97.8      30.3 Spring Groun~  
## 2 Barton Creek 01/08/2025 ~ Barton S~    -97.8      30.3 Spring Groun~  
## 3 Barton Creek 12/11/2024 ~ Old Mill~    -97.8      30.3 Spring Groun~  
## 4 Barton Creek 12/11/2024 ~ Barton S~    -97.8      30.3 Spring Groun~  
## 5 Barton Creek 12/11/2024 ~ Eliza Sp~    -97.8      30.3 Spring Groun~  
## 6 Barton Creek 11/21/2024 ~ Barton S~    -97.8      30.3 Spring Habit~  
## # i 15 more variables: PARAM_TYPE <chr>, PARAMETER <chr>, QUALIFIER <lg1>,  
## #   RESULT <dbl>, UNIT <chr>, FILTER <chr>, SAMPLE_ID <chr>,  
## #   SAMPLE_SITE_NO <dbl>, METHOD <chr>, QC_FLAG <chr>, PROJECT <chr>,  
## #   DATA_REF_NO <dbl>, SAMPLE_REF_NO <dbl>, TIME_NULL <chr>, QC_TYPE <lg1>
```

```
summary(water)
```

```
##   WATERSHED          SAMPLE_DATE          SITE_NAME          LON_DD_WGS84  
## Length:50068      Length:50068      Length:50068      Min.   :-97.77  
## Class :character  Class :character  Class :character  1st Qu.: -97.77  
##   LAT_DD_WGS84    SITE_TYPE          MEDIUM          PARAM_TYPE  
## Min.   :30.26    Length:50068      Length:50068      Length:50068  
## 1st Qu.:30.26    Class :character  Class :character  Class :character  
##   PARAMETER        QUALIFIER          RESULT          UNIT  
## Length:50068      Mode:logical    Min.   : -5.50    Length:50068  
## Class :character  NA's:50068      1st Qu.:  0.00    Class :character  
##   FILTER          SAMPLE_ID          SAMPLE_SITE_NO    METHOD  
## Length:50068      Length:50068      Min.   : 35.0     Length:50068  
## Class :character  Class :character  1st Qu.: 183.0     Class :character  
##   QC_FLAG          PROJECT          DATA_REF_NO      SAMPLE_REF_NO  
## Length:50068      Length:50068      Min.   : 91172     Min.   : 9986  
## Class :character  Class :character  1st Qu.:1692903    1st Qu.:305031  
##   TIME_NULL        QC_TYPE  
## Length:50068      Mode:logical  
## Class :character  NA's:50068  
## [ reached getOption("max.print") -- omitted 5 rows ]
```

2.3 Choosing 4 Sites for Analysis

```
# site options
```

```
unique(water$SITE_NAME)
```

```
## [1] "Barton Spring"
## [2] "Old Mill (Sunken Gardens) Spring"
## [3] "Eliza Spring"
## [4] "Old Mill Stream (outflow from Sunken Gardens Pool)"
## [5] "Upper Barton Spring"
## [6] "High Barton Spring"
```

Our analysis will focus on the Barton Spring, Eliza Spring, Old Mill (Sunken Gardens) Spring, and Upper Barton Spring sites.

2.4 Understanding Filter Fraction

```
unique(water$FILTER)
```

```
## [1] "Dissolved" "Total"      "N/A"
```

“Total” samples includes everything, from dissolved stuff to particles floating in the sample (suspended solids). “Dissolved” samples only include the dissolved substances, the sample is passed through a filter first, which removes solids. “N/A” is currently unknown.

2.5 Choosing Parameters to Review

The paramters variable is a representation of what measurement was taken for that row. Here, we display all available measurements for analysis in the dataset

```
# list of parameters
```

```
unique(water$PARAMETER)
```

```
## [1] "DISSOLVED OXYGEN"
## [2] "TOTAL TIME SPENT"
## [3] "NUMBER OF BARTON SPRINGS SALAMANDERS PHOTOGRAPHED"
## [4] "NUMBER OF AUSTIN BLIND SALAMANDERS PHOTOGRAPHED"
## [5] "FLOW"
## [6] "AUSTIN BLIND SLMNDR (1-2IN.)"
## [7] "AUSTIN BLIND SLMNDR (>=2IN.)"
## [8] "BARTON SPRINGS SLMNDR (TOTAL 1-2IN.)"
## [9] "BARTON SPRINGS SLMNDR (TOTAL >=2IN.)"
## [10] "BARTON SPRINGS SLMNDR (TOTAL <1IN.)"
## [11] "AUSTIN BLIND SLMNDR (<1IN.) (JUVENILE)"
## [12] "FLOW SEVERITY CODE (1=NONE;2=LOW;3=NORM;4=FLOOD;5=HIGH;6=DRY)"
## [13] "SUM OF BSS SALAMANDERS PER TRANSECT"
## [14] "SUM OF ABS SALAMANDERS PER TRANSECT"
## [15] "BASEFLOW"
## [16] "AUSTIN BLIND SALMNR (ADULT) > 1 INCH"
## [17] "BARTON SPGS SALMNR (ADULT) > 1 INCH"
## [18] "SALAMANDER (UNKNOWN LENGTH)"
## [19] "SALAMANDERS PHOTOGRAPHED >2 INCHES"
## [20] "SALAMANDERS PHOTOGRAPHED <1 INCH"
## [21] "SALAMANDERS PHOTOGRAPHED 1-2 INCHES"
## [22] "SALAMANDERS NOT PHOTOGRAPHED >2 INCHES"
## [23] "SALAMANDERS NOT PHOTOGRAPHED <1 INCH"
## [24] "SALAMANDERS NOT PHOTOGRAPHED 1-2 INCHES"
## [25] "AUSTIN BLIND SALAMANDER (UNKNOWN LENGTH)"
## [26] "BARTON SPRINGS SLMNDR (1-2IN.) IN COBBLE/BOULDER"
## [27] "BARTON SPRINGS SLMNDR (>=2IN.) IN COBBLE/BOULDER"
## [28] "BARTON SPRINGS SLMNDR (1-2IN.) IN VEGETATION/MOSS"
```

```
## [29] "BARTON SPRINGS SLMNDR (>=2IN.) IN VEGETATION/MOSS"
## [30] "AUSTIN BLIND SLMNDR (1-2IN.) ON EXPOSED SUBSTRATE"
## [31] "AUSTIN BLIND SLMNDR (>=2IN.) ON EXPOSED SUBSTRATE"
## [32] "AUSTIN BLIND SLMNDR (1-2IN.) IN LEAF LITTER"
## [33] "AUSTIN BLIND SLMNDR (>=2IN.) IN LEAF LITTER"
## [34] "AUSTIN BLIND SLMNDR (>=2IN.) IN COBBLE/BOULDER"
## [35] "AUSTIN BLIND SLMNDR (1-2IN.) IN COBBLE/BOULDER"
## [36] "AUSTIN BLIND SLMNDR (1- 2IN.) IN VEGETATION/MOSS"
## [37] "AUSTIN BLIND SLMNDR (>=2IN.) IN VEGETATION/MOSS"
## [38] "BARTON SPRINGS SLMNDR (<1IN.) ON EXPOSED SUBSTRATE"
## [39] "BARTON SPRINGS SLMNDR (<1IN.) IN LEAF LITTER"
## [40] "BARTON SPRINGS SLMNDR (<1IN.) IN COBBLE/BOULDER"
## [41] "AUSTIN BLIND SLMNDR (<1IN.) ON EXPOSED SUBSTRATE"
## [42] "AUSTIN BLIND SLMNDR (<1IN.) IN LEAF LITTER"
## [43] "AUSTIN BLIND SLMNDR (<1IN.) IN COBBLE/BOULDER"
## [44] "AUSTIN BLIND SLMNDR (<1IN.) IN VEGETATION/MOSS"
## [45] "BARTON SPRINGS SLMNDR (<1IN.) IN VEGETATION/MOSS"
## [46] "BARTON SPRINGS SLMNDR (>=2IN.) ON EXPOSED SUBSTRATE"
## [47] "BARTON SPRINGS SLMNDR (1-2IN.) ON EXPOSED SUBSTRATE"
## [48] "BARTON SPRINGS SLMNDR (1-2IN.) IN LEAF LITTER"
## [49] "BARTON SPRINGS SLMNDR (>=2IN.) IN LEAF LITTER"
## [50] "AREA SAMPLED"
## [ reached getOption("max.print") -- omitted 12 entries ]
```

The definitions of these were provided by Mr. Robert Clayton from the City of Austin Watershed Protection Department. For this analysis, the measurements of interest include DISSOLVED OXYGEN, FLOW SEVERITY CODE (1=NONE;2=LOW;3=NORM;4=FLOOD;5=HIGH;6=DRY), NUMBER OF AUSTIN BLIND SALAMANDERS PHOTOGRAPHED, and AREA SAMPLED.

2.6 Associated Units List

The units of interest all correspond to the measurements of interest, so we removed all unnecessary unit types from the dataset for analysis

```
unique(water$UNIT)
```

```
## [1] "MG/L"
## [2] "Minutes"
## [3] "Count"
## [4] "Cubic Feet Per Second"
## [5] "None"
## [6] "1-2: (Yes; No)"
## [7] "Feet per second"
## [8] "(%) Percent"
## [9] "Sq Ft"
## [10] "Inches"
## [11] "Feet"
## [12] "MM"
## [13] "gal/min"
## [14] "Relative Abundance (1): <20 (2): 21-50 (3): >50"
```

The units of interest are “MG/L”, “Sq Ft”, “None”, and “Count”. “None” is there for the flowcode category.

3 Data Cleaning

The raw dataset stores each measurement as its own row, with the specific measurement type recorded in `PARAMETER` and its value in `RESULT`. This makes metrics from a single site visit appear as different observations (rows). To analyze relationships between these variables, the data needs to be reshaped so that each row represents a single site visit, with each measurement type as its own column.

Several complications arose in this reshaping process, worth noting explicitly:

- 1.) Some site visits included multiple sub-samples (i.e measurements taken at different depths or along different survey transects on the same day). Rather than treating these as separate observations (which would fragment a single visit across multiple rows) or arbitrarily keeping only one, replicate values were averaged together to produce a single representative value per site per day.
- 2.) The original sample date included exact time of day, and different measurements taken during the same visit were sometimes logged at slightly different timestamps. Grouping by the full timestamp caused measurements from the same visit to be split into separate rows. This was resolved by grouping on the calendar date alone, which correctly reunites same-day measurements.
- 3.) An earlier version of the cleaning step used to filter for site names used `==` to filter to the four sites of interest, which unintentionally checks values by position rather than membership, silently dropping some valid rows. This was corrected using `%in%`, which properly matches each row against the full list of target sites regardless of order.
- 4.) Some Dissolved Oxygen readings were recorded in “% Saturation” rather than mg/L, alongside a separate, correctly-scaled mg/L reading for the same site visit. Since these two measurements represent fundamentally different scales, averaging them together during pivoting produced physically implausible DO values (e.g., 60+ mg/L). This was resolved by *excluding percent-saturation readings* before reshaping, ensuring *only mg/L values were averaged into the DO column*.

After reviewing the original dataset and what is important for this analysis, the following is the cleaning done to achieve the desired new dataset to work with.

```
# cleaned data, all managed in one spot and edited throughout the analysis
water_cleaned <- water |>
  # exclude percent-saturation readings
  filter(!(PARAMETER == "DISSOLVED OXYGEN" & UNIT == "(%) Percent")) |>
  # uniform date format
  mutate(sample_day = as_date(mdy_hms(SAMPLE_DATE))) |>
  pivot_wider(
    id_cols = c(sample_day, SITE_NAME),
    # the values of the variable `parameter` will become variables
    names_from = PARAMETER,
    # the cell values of `result` will match the corresponding variable
    values_from = RESULT,
    # averages replicate sub-samples
    values_fn = ~ mean(.x, na.rm = TRUE)
  ) |>
  # selecting variables for analysis and changing names
  select(
    sampledate = sample_day,
    site = SITE_NAME,
    DO = `DISSOLVED OXYGEN`,
    Flowcode = "FLOW SEVERITY CODE (1=NONE;2=LOW;3=NORM;4=FLOOD;5=HIGH;6=DRY)",
    salamanders = "NUMBER OF AUSTIN BLIND SALAMANDERS PHOTOGRAPHED",
    area = "AREA SAMPLED",
```

```

) |>
# choosing specific sites
filter(site %in% c("Barton Spring",
                  "Eliza Spring",
                  "Old Mill (Sunken Gardens) Spring",
                  "Upper Barton Spring"))

head(water_cleaned)

## # A tibble: 6 x 6
##   sampled date site                DO Flowcode salamanders area
##   <date>      <chr>                <dbl>   <dbl>      <dbl> <dbl>
## 1 2025-02-25 Barton Spring          4.61     NA         NA     NA
## 2 2025-01-08 Barton Spring          4.29     NA         NA     NA
## 3 2024-12-11 Old Mill (Sunken Gardens) Spring 2.63     NA         NA     NA
## 4 2024-12-11 Barton Spring          4.18     NA         NA     NA
## 5 2024-12-11 Eliza Spring           4.36     NA         NA     NA
## 6 2024-11-21 Barton Spring          NA       NA          0     NA

```

After cleaning, the dataset contains site-visit-level observations for the four springs of interest, with dissolved oxygen, flow severity, salamander counts, and survey area as separate columns. It's worth noting that DO is measured far more frequently and consistently than salamander counts or area surveys, which appear to follow a more occasional or opportunistic sampling schedule. This is reflected in the higher proportion of missing values for those two variables. This is a limitation to keep in mind when interpreting any results involving salamander presence or survey area.

4 Results

4.1 Summary Stats

Before comparing DO across sites, it's worth first checking the overall distribution of DO values and how completely it was recorded at each site. This establishes a baseline understanding of the variable itself, and confirms that missing data isn't concentrated at any one site in a way that could bias later comparisons.

```

summary(water_cleaned$DO)

##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##  0.000   5.140   5.970   5.945   6.670  21.470   893

```

```

water_cleaned |>
  group_by(site) |>
  summarise(n = n(),
            n_missing_DO = sum(is.na(DO))
            )

```

```

## # A tibble: 4 x 3
##   site                n n_missing_DO
##   <chr>            <int>      <int>
## 1 Barton Spring      964        343
## 2 Eliza Spring       687        227
## 3 Old Mill (Sunken Gardens) Spring 542        222
## 4 Upper Barton Spring 319        101

```

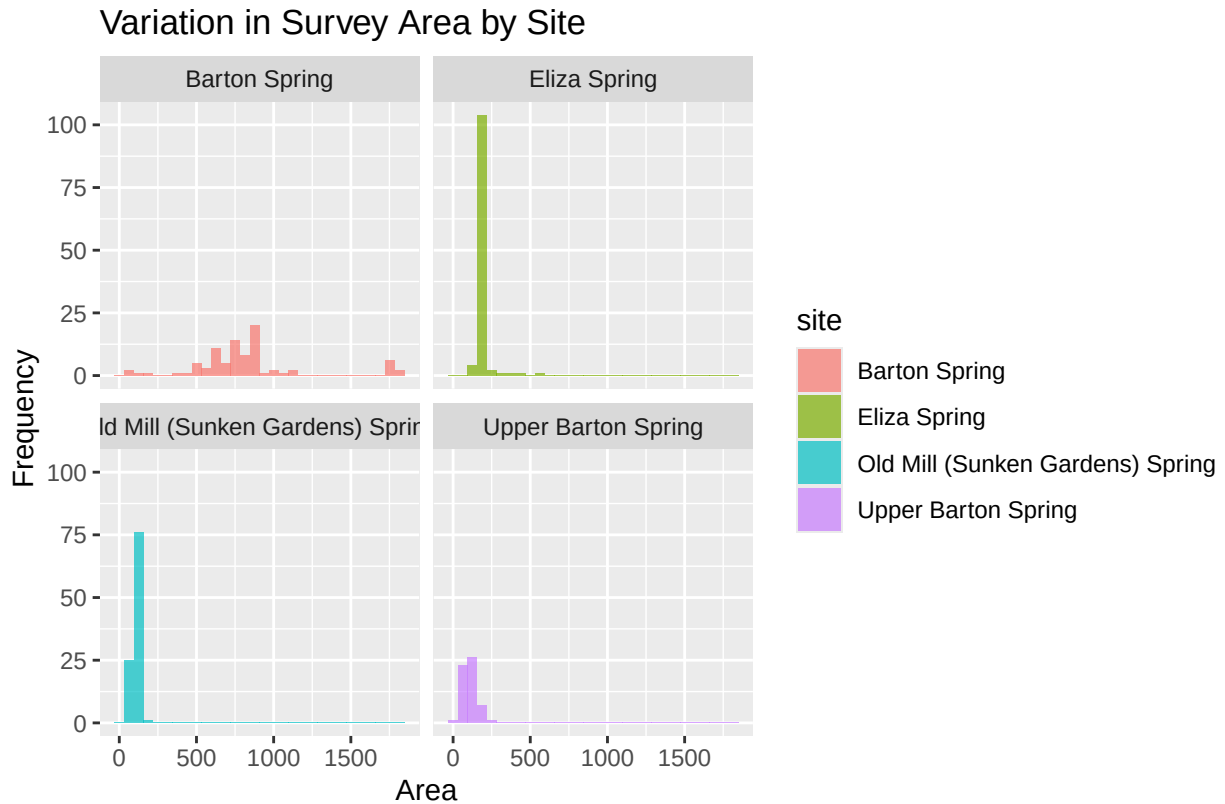
Missing DO readings are broadly similar in proportion across all four sites (32-41%), suggesting no single site's DO comparison is disproportionately affected by data availability.

4.2 Survey Area by Site

Before directly addressing how dissolved oxygen (DO) levels vary across different spring sites, and what is their relationship with seasonal changes is, it is important to know some things about the sites themselves. First, we looked at what the variation in areas sampled for each site looked like

```
water_cleaned |>
  ggplot() +
  geom_histogram(aes(x = area, fill = site), alpha = 0.7) +
  facet_wrap(~site) +
  labs(title = "Variation in Survey Area by Site",
       x = "Area",
       y = "Frequency",
       caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

```
## Warning: Removed 2152 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



Data from the City of Austin Barton Springs Salamanders DO and Flow

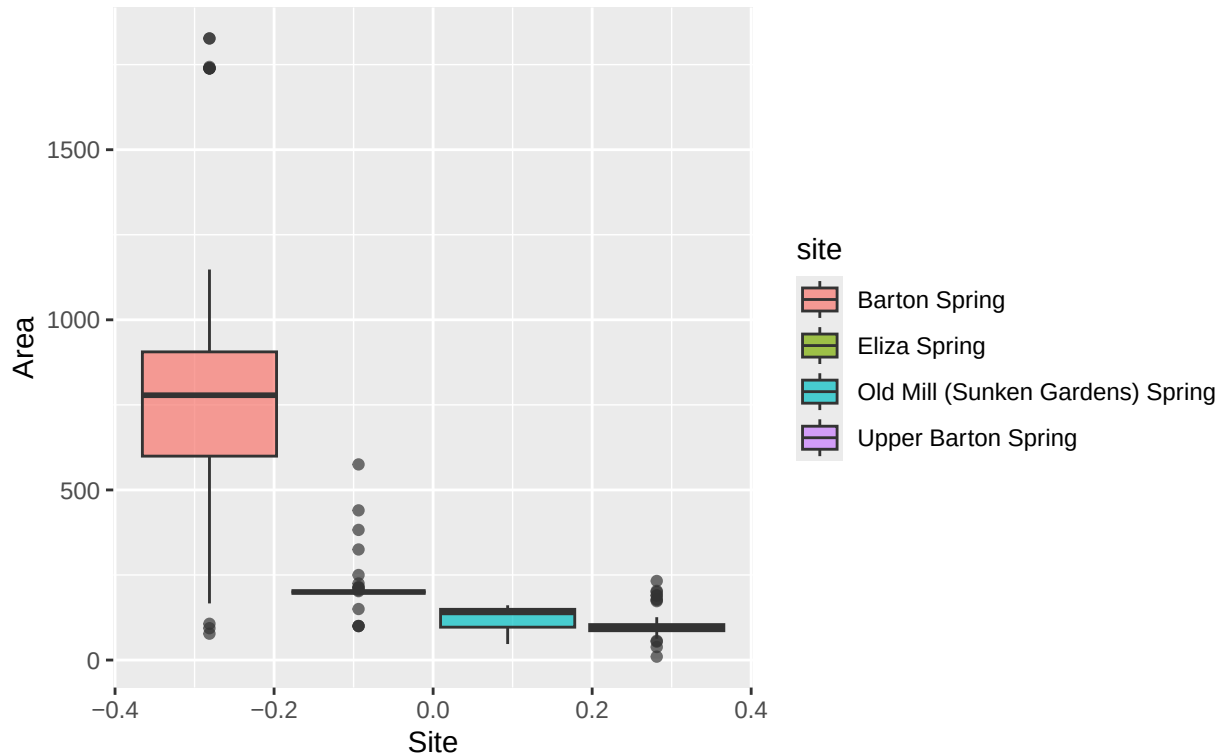
```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_boxplot(aes(y = area, fill = site), alpha = 0.7) +
  labs(title = "Variation in Survey Area by Site",
       x = "Site",
       y = "Area",
```



```
caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

```
## Warning: Removed 2152 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

Variation in Survey Area by Site



Data from the City of Austin Barton Springs Salamanders DO and Flow

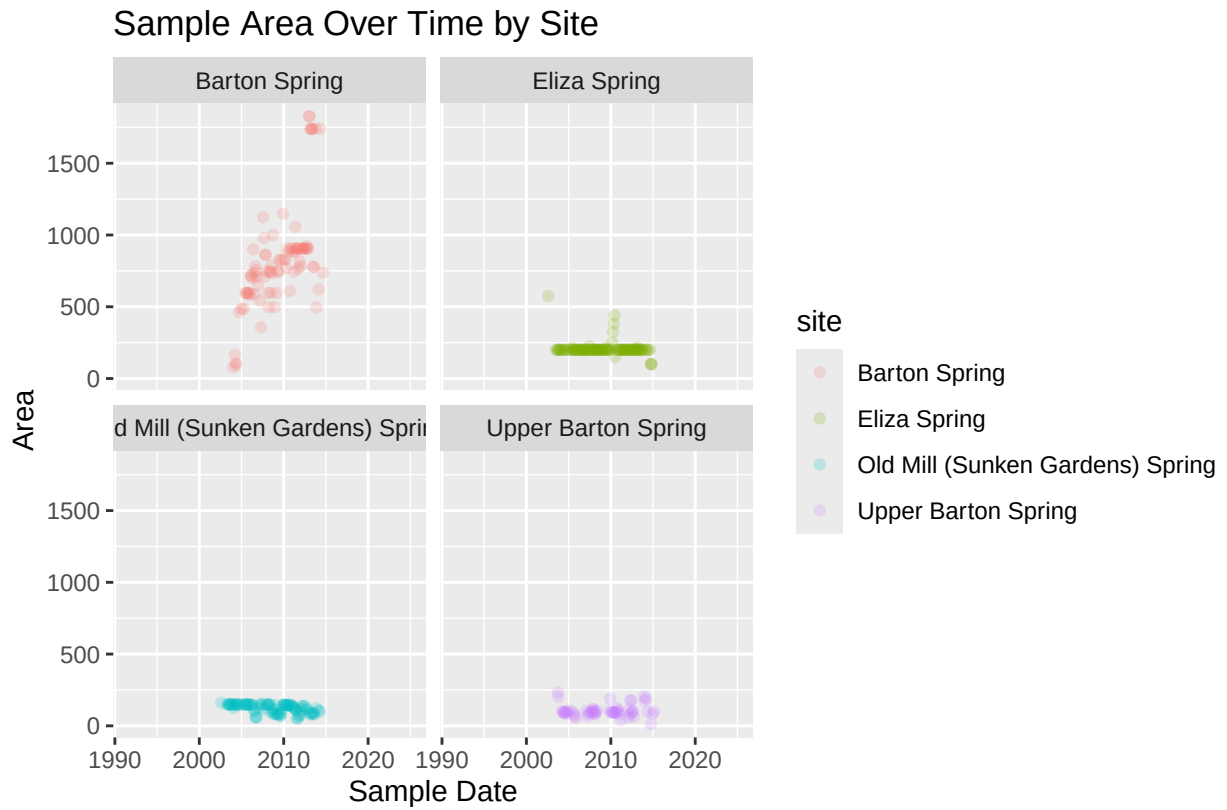
While the areas sampled for Eliza, Old Mill (Sunken Garden), and Upper Barton Springs were relatively consistent, Barton Spring's sample areas were much more spread, and also had the greatest outlier among them all. This raises an important question to keep in mind going forward: if Barton Spring is sampled over a much larger and more variable area than the other sites, any observed differences in DO or salamander detection at Barton Spring may partly reflect differences in survey effort rather than the site itself

4.3 Sample Area Over Time

We were then interested in the when these surveys took place, and if anything could be noticed to that helps explain the Barton Springs outliers

```
water_cleaned |>
  ggplot() +
  geom_point(aes(x = sampleddate, y = area, color = site), alpha = 0.2) +
  facet_wrap(~site) +
  labs(title = "Sample Area Over Time by Site",
        x = "Sample Date",
        y = "Area",
        caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

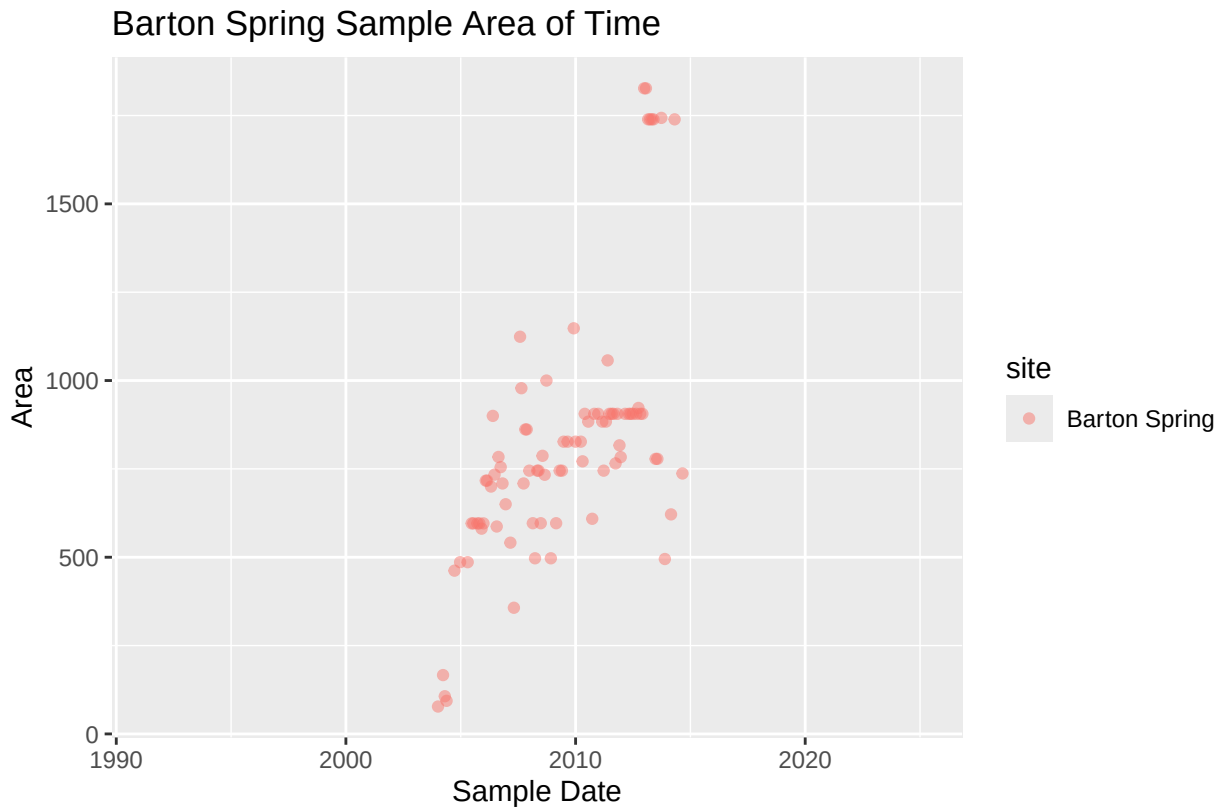
```
## Warning: Removed 2152 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Data from the City of Austin Barton Springs Salamanders DO and Flow

```
water_cleaned |>
  filter(site == "Barton Spring") |>
  ggplot() +
  geom_point(aes(x = sampleddate, y = area, color = site), alpha = 0.5) +
  labs(title = "Barton Spring Sample Area of Time",
        x = "Sample Date",
        y = "Area",
        caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

Warning: Removed 878 rows containing missing values or values outside the scale range
(`geom\_point()`).



The Barton Spring outliers are not randomly distributed in time. All three occur within a narrow window around 2013, distinctly separated from the site's typical survey area range (roughly 100-1150 sq ft) seen across the rest of the 2004-2015 sampling period. This suggests a specific, likely temporary change in survey protocol or scope at that site around 2013, rather than ongoing variability. Without further documentation from the data source, the exact cause of this change can't be determined here. It is worth flagging as an anomaly isolated to a specific period rather than a general characteristic of Barton Spring sampling.

Taking a closer look at the 3 highest area surveys for Barton Spring

```
# Check high values of area
water_cleaned |>
  slice_max(n = 3, area)
```

```
## # A tibble: 3 x 6
##   sampledte site      DO Flowcode salamanders area
##   <date>      <chr>    <dbl>    <dbl>      <dbl> <dbl>
## 1 2013-01-24 Barton Spring 5.05      NA        NA 1827
## 2 2012-12-27 Barton Spring 5.1       NA        NA 1827
## 3 2013-09-26 Barton Spring 4.8       NA        NA 1743.
```

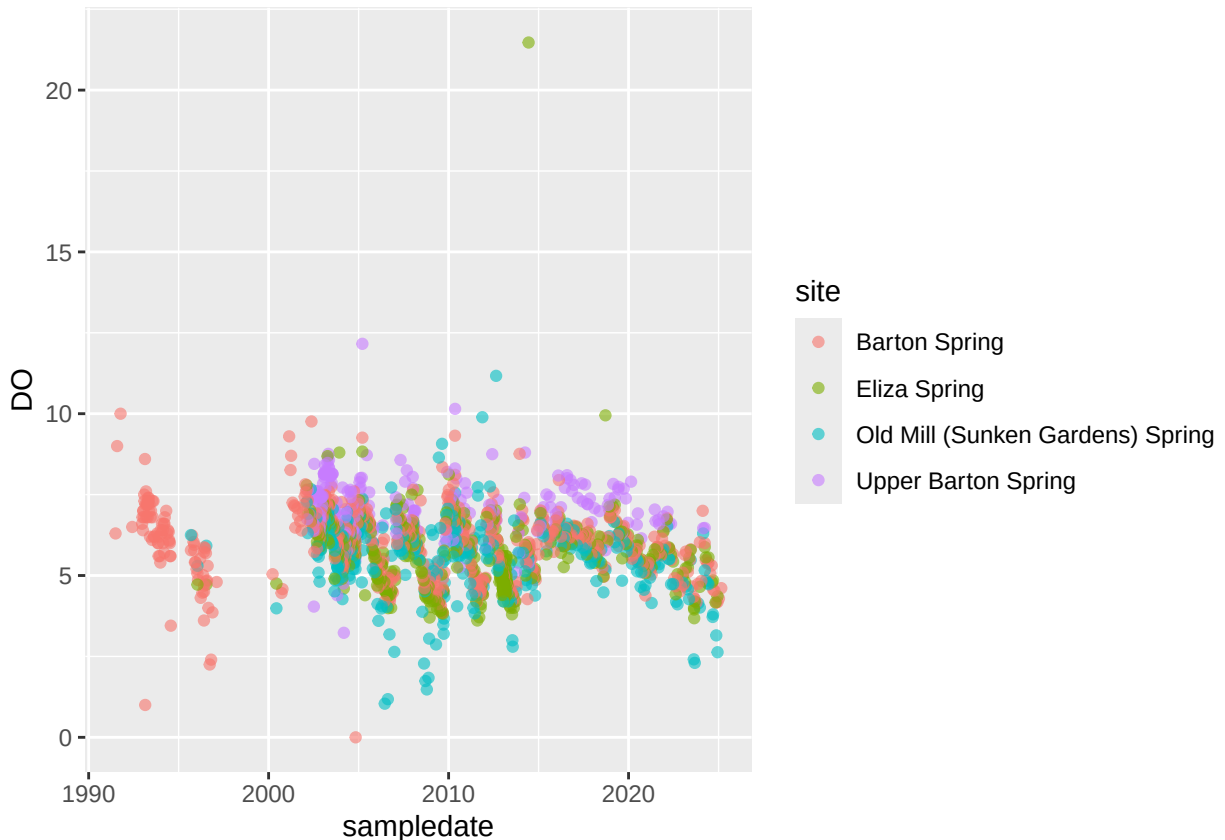
The highest survey area for Barton Spring is approximately 1827 Sq Ft, taking place on two occasions in late 2012/early 2013, with a third closely comparable outlier (1743 Sq Ft) later that year.

4.4 DO Over Time

With the exception of Barton Spring, most of the sites have consistent survey areas. How do their dissolved oxygen levels vary over time?

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_point(aes(x = sampleddate, y = DO, color = site), alpha = 0.6)
```

```
## Warning: Removed 893 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Given how unusual this value is, it's worth verifying the units are correctly recorded and that this isn't an artifact of the sub-sample averaging step, rather than a genuine reading.

```
water |>
  filter(PARAMETER == "DISSOLVED OXYGEN",
         SITE_NAME == "Eliza Spring",
         str_detect(SAMPLE_DATE, "2014")) |>
  select(SAMPLE_DATE, SITE_NAME, RESULT, UNIT, FILTER)
```

```
## # A tibble: 12 x 5
##   SAMPLE_DATE      SITE_NAME  RESULT UNIT  FILTER
##   <chr>          <chr>    <dbl> <chr> <chr>
## 1 10/22/2014 12:35:00 PM Eliza Spring  4.93 MG/L Dissolved
## 2 10/15/2014 09:58:00 AM Eliza Spring  5.21 MG/L Dissolved
## 3 10/14/2014 10:49:00 AM Eliza Spring  5.14 MG/L Dissolved
## 4 08/14/2014 09:30:00 AM Eliza Spring  5.24 MG/L Dissolved
## 5 08/14/2014 09:30:00 AM Eliza Spring  4.6 MG/L Dissolved
## 6 06/12/2014 09:44:00 AM Eliza Spring 21.6 MG/L Dissolved
## 7 06/12/2014 09:44:00 AM Eliza Spring 21.3 MG/L Dissolved
## 8 06/11/2014 11:35:00 AM Eliza Spring  5.7 MG/L Dissolved
```

```
## 9 04/24/2014 08:15:00 AM Eliza Spring 5.86 MG/L Dissolved
## 10 04/10/2014 09:30:00 AM Eliza Spring 5 MG/L Dissolved
## 11 04/01/2014 01:10:00 PM Eliza Spring 5.95 MG/L Dissolved
## 12 02/13/2014 10:50:00 AM Eliza Spring 6.4 MG/L Dissolved
```

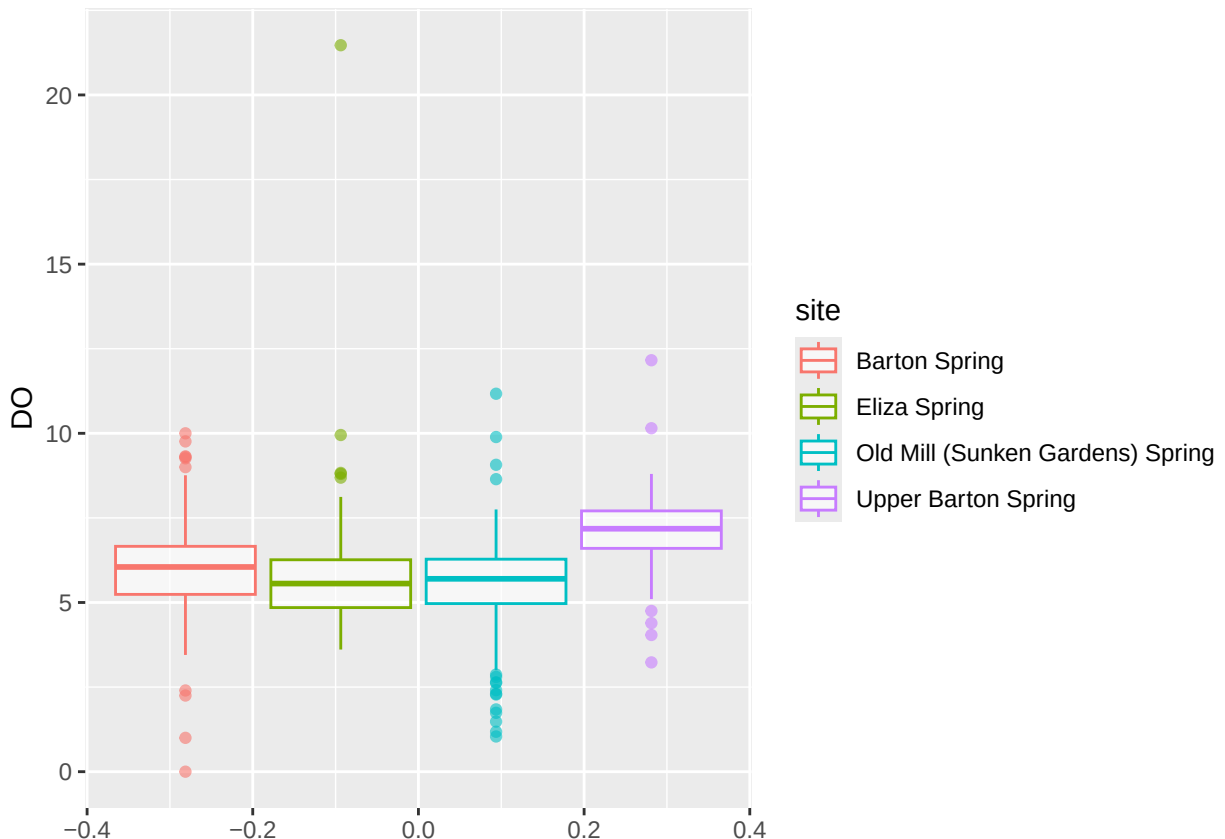
One elevated DO reading at Eliza Spring in June 2014 (~21.5 mg/L, confirmed across two sub-samples on the same visit) stands out from the otherwise consistent 0-13 mg/L range seen throughout the dataset. Unlike the percent-saturation contamination addressed above, this value is correctly recorded in mg/L and is retained as a genuine observation, possibly reflecting a short-term supersaturation event (e.g. algal activity or aeration) rather than a data quality issue. This would be an excellent question for city officials who conducted the survey.

DO levels appear broadly consistent over time across all sites, with the single exception of the Eliza Spring outlier noted above. This raises the question of how DO varies by site specifically.

4.5 DO by Site

```
# log10(DO) by site
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_boxplot(aes(y = DO, color = site), alpha = 0.6)
```

```
## Warning: Removed 893 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



DO levels are broadly similar in typical range across the four sites (median values cluster between roughly 5.5 and 7.5 mg/L), but with some notable site-specific patterns. Upper Barton

Spring shows the highest typical DO of the four sites. Old Mill (Sunken Gardens) Spring has the tightest core distribution but includes several low-DO outliers, pulling down its overall range. Barton Spring shows the widest spread, including a few near-zero readings. Eliza Spring's single elevated outlier (~21.5 mg/L, discussed above) stands apart from an otherwise typical distribution.

4.6 Salamanders

```
water_cleaned |>
  summarize("Total salamander count" = sum(salamanders, na.rm = T))
```

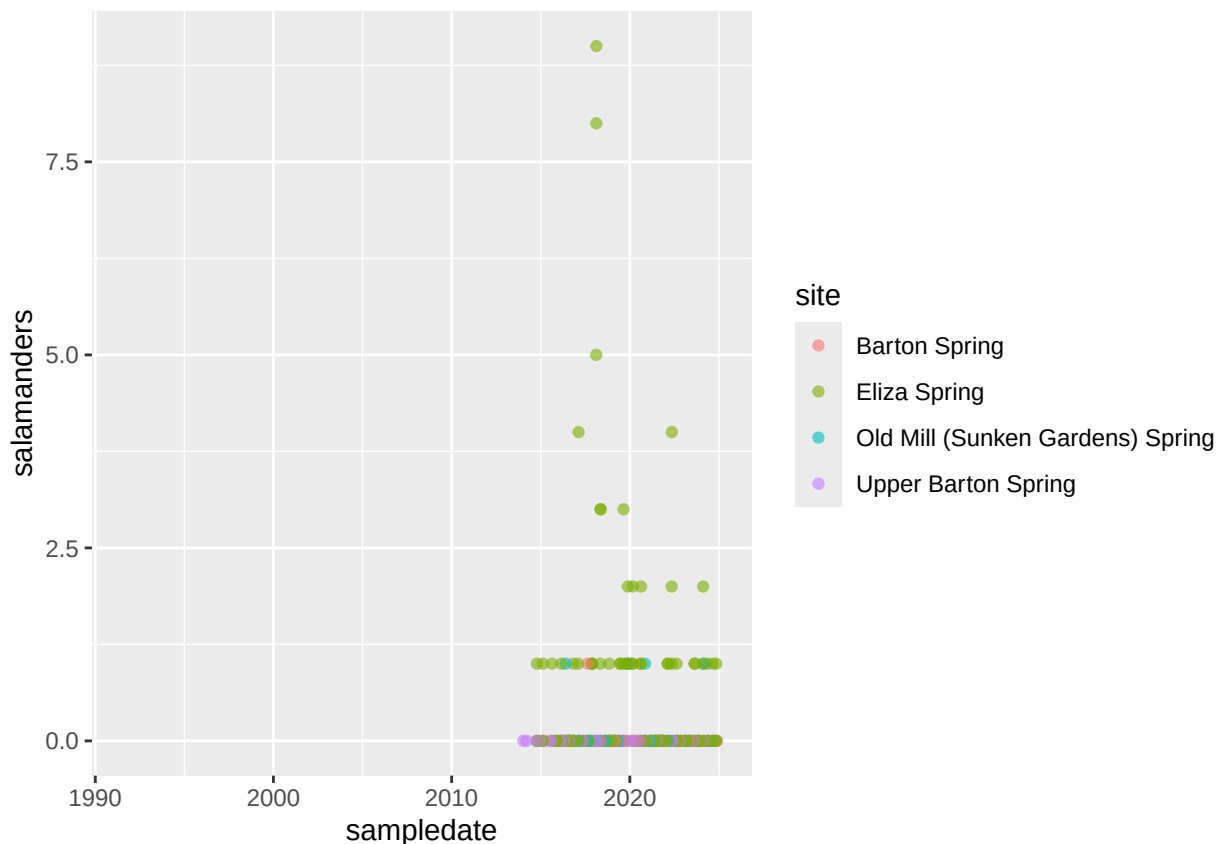
```
## # A tibble: 1 x 1
##   `Total salamander count`
##               <dbl>
## 1                   84
```

A total of 84 salamanders were counted across the entire dataset. This is not to say that there exist 84 salamanders in total though, as several are likely repeated counts over time.

How is this distributed by region?

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_point(aes(x = sampledate, y = salamanders, color = site), alpha = 0.6)
```

```
## Warning: Removed 2297 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



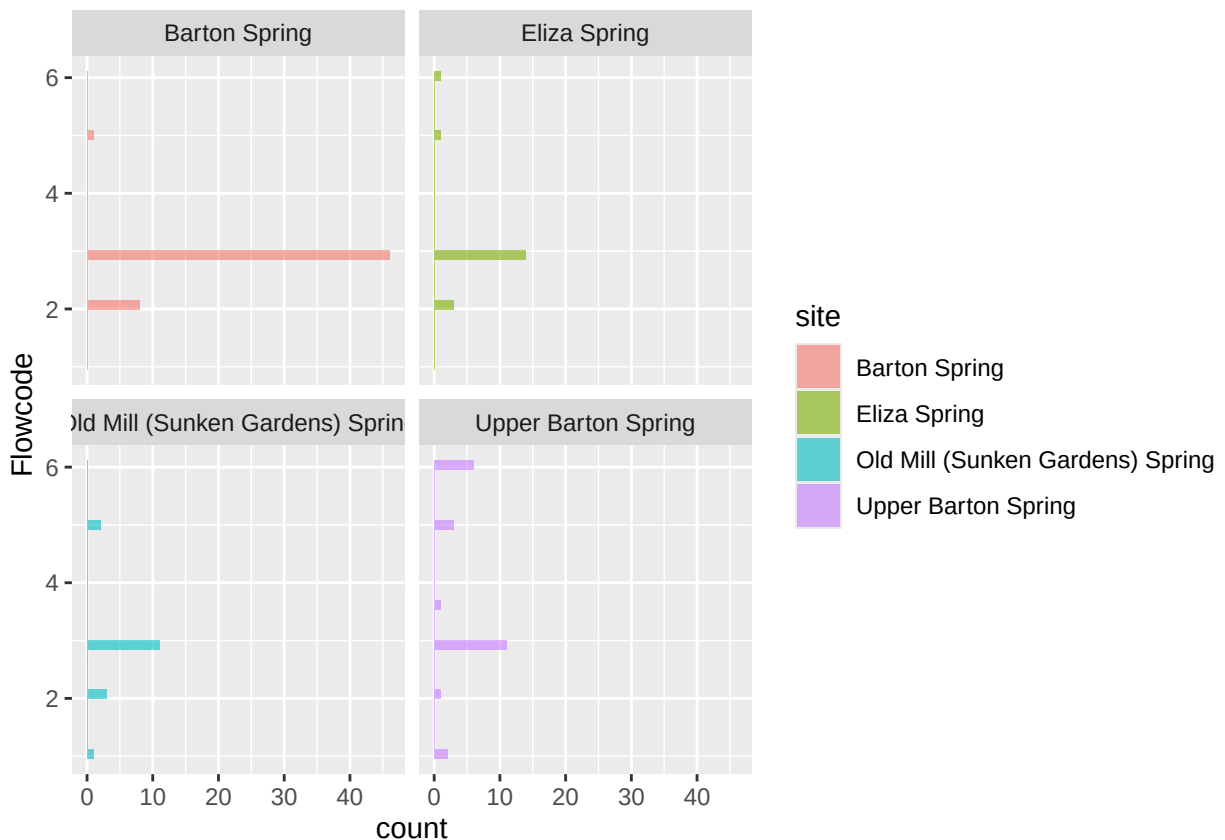
Seems like the salamander counting for this view does not start until around 2015, or that there were no salamanders to count until then, and that they like Eliza Spring to most. Regardless, Eliza Spring's markedly higher salamander detection rate, combined with Barton Spring's earlier-noted variability in survey area, suggests that site-level differences in either measured DO or detected salamanders may partly reflect differences in sampling practices rather than the underlying spring conditions themselves.

4.7 Flowcodes

Given this is the case, Does flow perhaps effect DO?

```
water_cleaned |>
  ggplot() +
  geom_histogram(aes(y = Flowcode, fill = site), alpha = 0.6) +
  facet_wrap(~site)
```

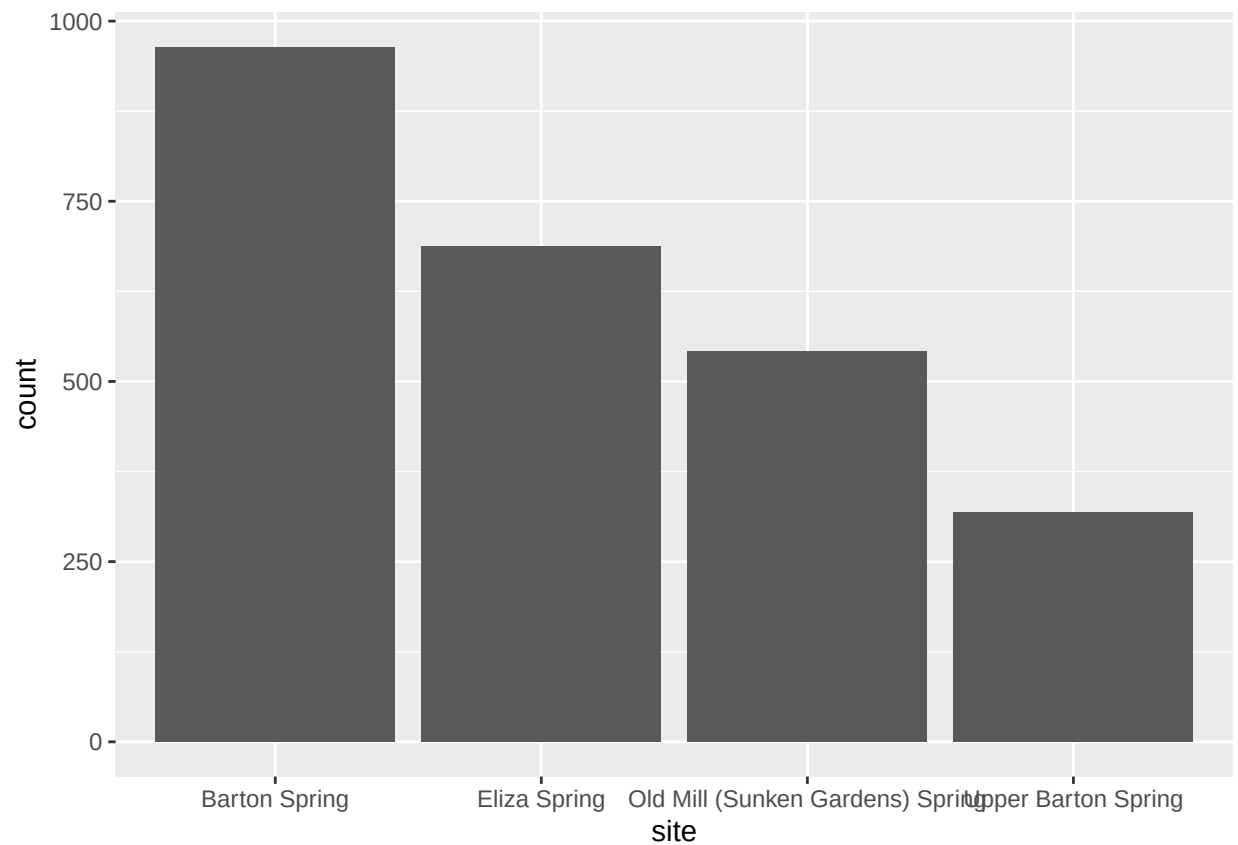
```
## Warning: Removed 2397 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



This suggests that Barton Spring and Eliza Spring are somewhat over-represented in the dataset. Let's check for sampling country for each

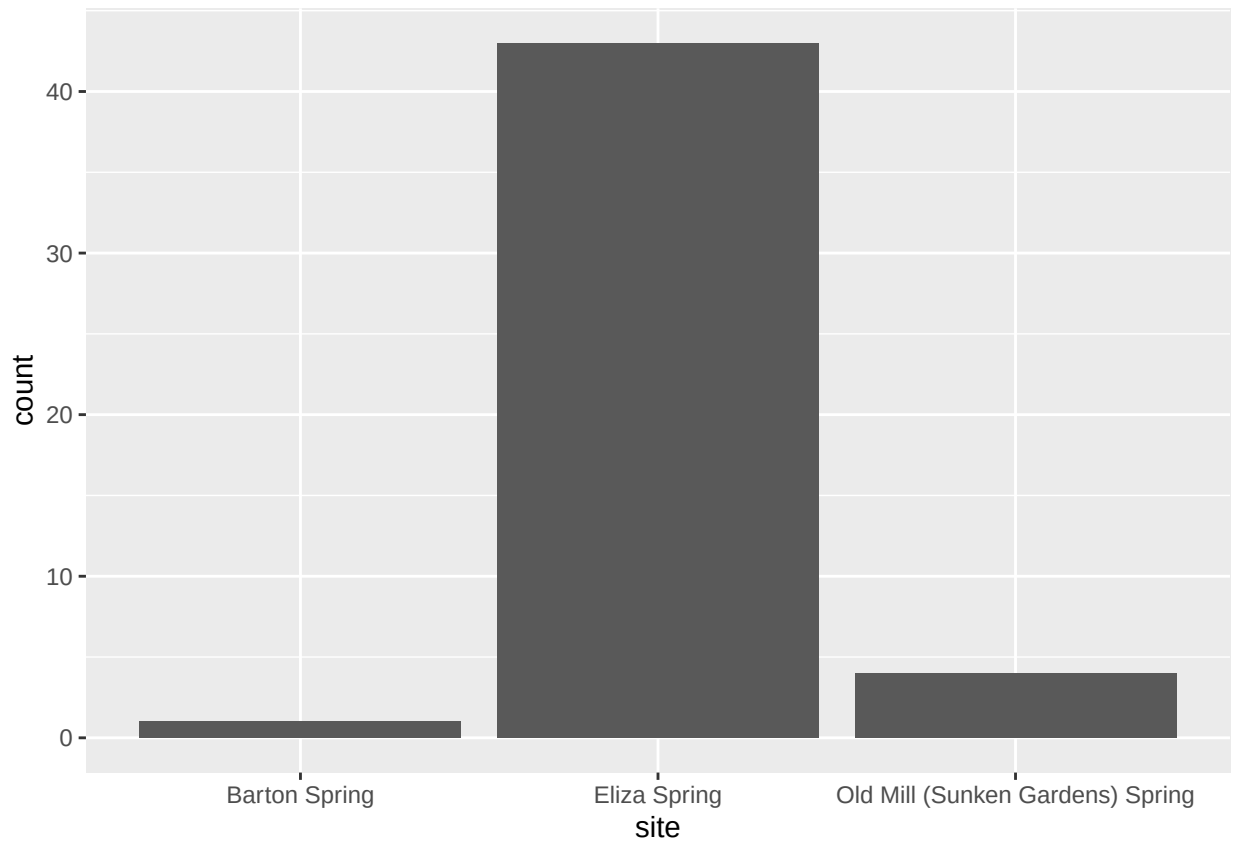
4.8 Inquiries Before Modeling

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_bar(aes(x = site))
```



And wouldn't you believe it, they are indeed the most sampled sites from this view. Lets check where salamanders are being looked for the most

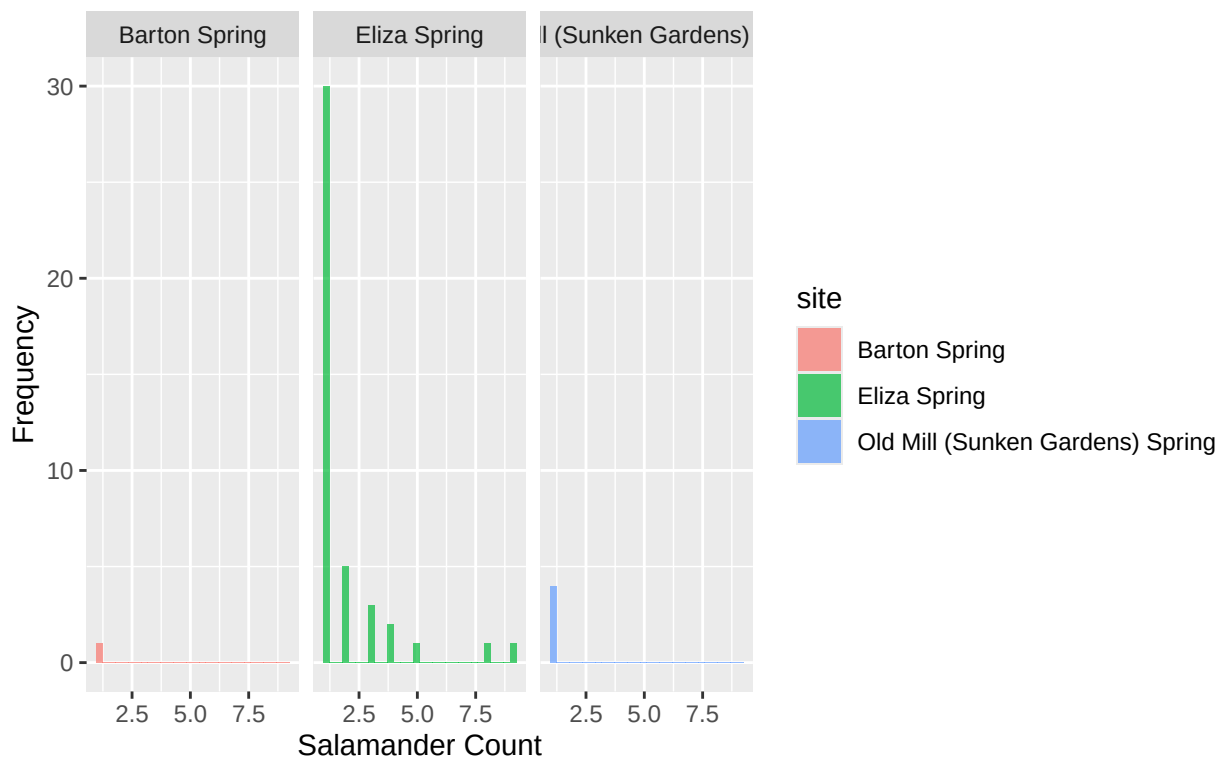
```
water_cleaned |>
  group_by(site) |>
  filter(salamanders != 0) |>
  ggplot() +
  geom_bar(aes(x = site))
```

Salamander detections were recorded at three of the four sites. Eliza Spring far more frequently than the others (over 40 site-visits with at least one detection), Old Mill less often, and Barton Spring with a single incidental detection. Upper Barton Spring had no recorded detections at all

```
water_cleaned |>
  filter(salamanders != 0) |>
  ggplot() +
  geom_histogram(aes(x = salamanders, fill = site), alpha = 0.7) +
  facet_wrap(~site) +
  labs(title = "Salamander Count Distributions by Spring",
       x = "Salamander Count",
       y = "Frequency",
       caption = "Data from the City of Austin Barton Springs Salamanders D0 and Flow")
```

Salamander Count Distributions by Spring



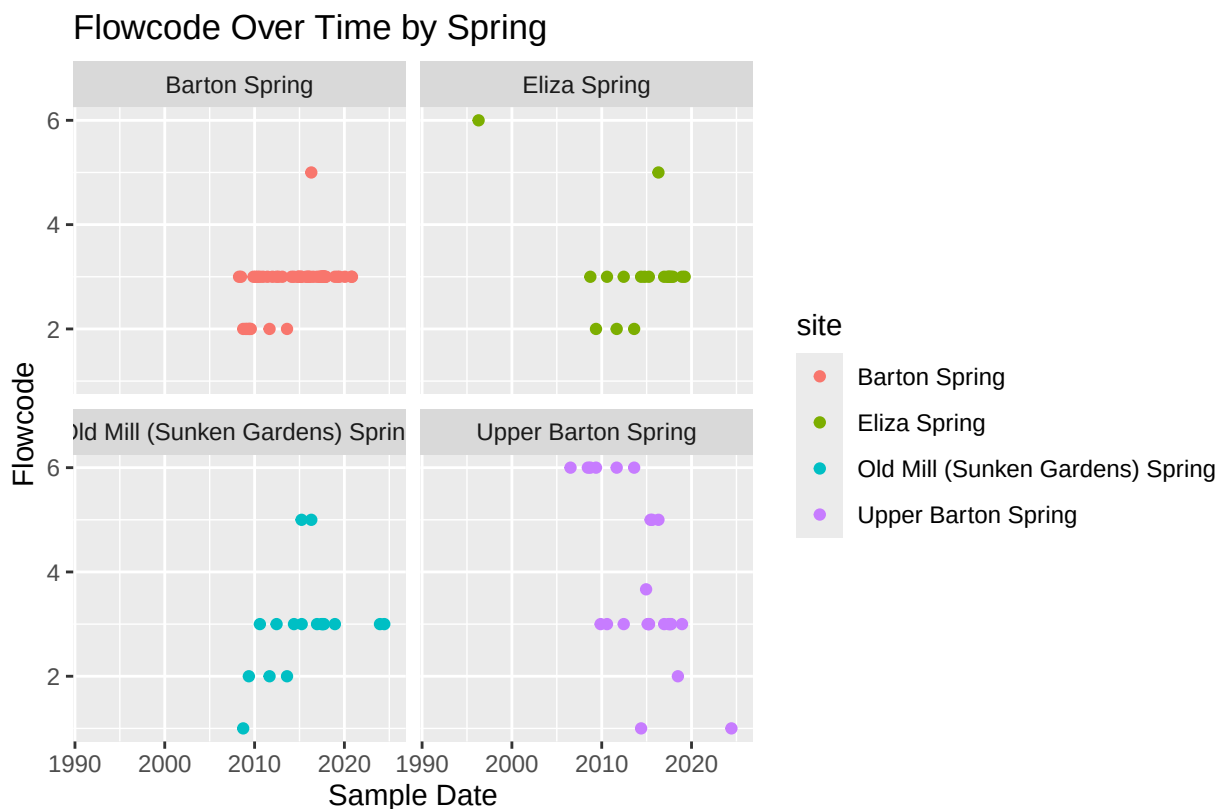
Data from the City of Austin Barton Springs Salamanders DO and Flow

Among the sites with any detections, the shape of the counts themselves also differs. Eliza Spring not only has far more site-visits with a detection, but also accounts for nearly all of the higher counts, while Barton Spring and Old Mill (Sunken Gardens) Spring are limited to single-digit low counts. This reinforces the earlier point that Eliza Spring is functionally the “salamander site” in this dataset, both in how often they’re found there and how many are found at once. This is worth keeping in mind as a confound when comparing DO or flow patterns across sites.

Having looked at how DO, survey area, and salamander detections each vary across sites, the last piece of the confounding picture is flow itself. Since flow conditions can influence both DO and salamander activity, it’s worth checking whether the four sites were even sampled under comparable flow regimes to begin with.

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_point(aes(x = sampleddate, y = Flowcode, color = site)) +
  facet_wrap(~site) +
  labs(title = "Flowcode Over Time by Spring",
       x = "Sample Date",
       y = "Flowcode",
       caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

```
## Warning: Removed 2397 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



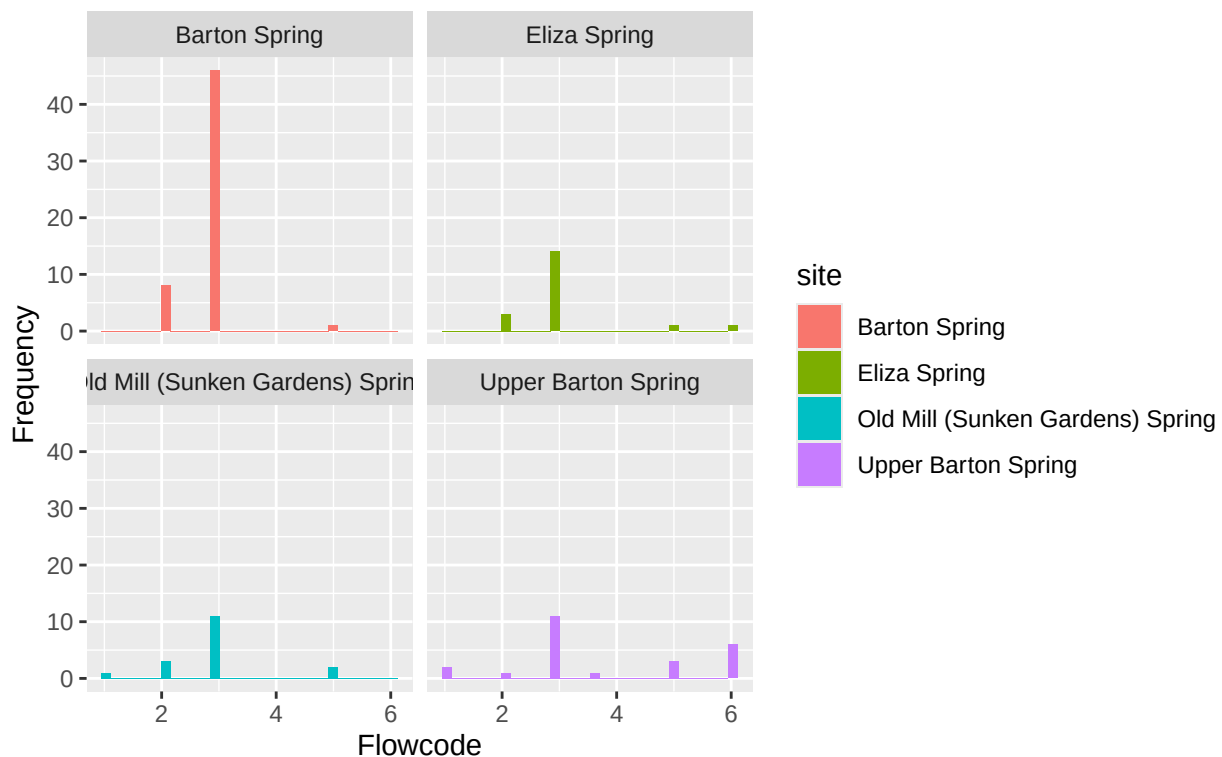
Data from the City of Austin Barton Springs Salamanders DO and Flow

Looking at flow conditions over time rather than just their overall frequency, Barton Spring's readings sit almost entirely at code 3 (NORM) across its full sampling window, with only a couple of LOW readings breaking that pattern. Eliza Spring, Old Mill, and Upper Barton Spring each show more scatter across the flow scale throughout their sampling periods, including a mix of LOW, NORM, and HIGH/DRY readings recorded within the same few years. There's no clear indication that any site was sampled disproportionately during a specific flow regime relative to its own history. The spread looks like normal seasonal or year-to-year variation rather than a shift in sampling protocol.

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_histogram(aes(x = Flowcode, fill = site)) +
  facet_wrap(~site) +
  labs(title = "Flowcode Distributions by Spring",
       x = "Flowcode",
       y = "Frequency",
       caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

```
## Warning: Removed 2397 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

Flowcode Distributions by Spring



Data from the City of Austin Barton Springs Salamanders DO and Flow

Looking at the overall shape of these distributions confirms the pattern: Barton Spring's flow readings are heavily concentrated at code 3 (NORM), with a much smaller cluster at code 2 (LOW) and almost nothing else. Eliza Spring, Old Mill (Sunken Gardens) Spring, and Upper Barton Spring all show a similar peak at NORM but with more mass spread across LOW, FLOOD, HIGH, and DRY codes. In short, all four sites were sampled predominantly under normal flow conditions, which is reassuring for comparing DO across sites. However, Barton Spring in particular has very little representation at the flow extremes, so any site-level DO patterns tied to HIGH or DRY conditions are effectively only informed by the other three springs.

5 Modeling

Having looked at DO, survey area, salamander detection, and flow separately across sites, the remaining question is whether the site-level differences in DO seen earlier hold up once we account for the overall time trend, and whether they can be explained away by survey effort or salamander detection rates. We start with a baseline model regressing DO on sampled date and site alone, which tells us whether site is a significant predictor of DO once time is controlled for. We'll then attempt to extend this model with area and salamanders as additional predictors, to test directly whether the site effect shrinks once those two variables are accounted for.

```
# Getting rid of NA values
model_water <- water_cleaned |>
  filter(!is.na(DO))
# Fit model
lin_model <- lm(DO ~ sampleddate + site, data = model_water)
summary(lin_model)
```

```
##
## Call:
```

```
## lm(formula = DO ~ sampleddate + site, data = model_water)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.0723 -0.6338  0.0705  0.6427 15.9484
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      7.105e+00  1.548e-01  45.901 < 2e-16 ***
## sampleddate     -8.115e-05  1.056e-05  -7.685 2.64e-14 ***
## siteEliza Spring -2.659e-01  6.837e-02  -3.890 0.000104 ***
## siteOld Mill (Sunken Gardens) Spring -3.589e-01  7.605e-02  -4.719 2.58e-06 ***
## siteUpper Barton Spring      1.207e+00  8.657e-02  13.944 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.099 on 1614 degrees of freedom
## Multiple R-squared:  0.2008, Adjusted R-squared:  0.1989
## F-statistic: 101.4 on 4 and 1614 DF,  p-value: < 2.2e-16
```

Site is a highly significant predictor of DO even after controlling for the overall downward trend over time (sampledate: -8.1×10^{-5} mg/L per day, $p < 2 \times 10^{-16}$). Relative to Barton Spring, Eliza Spring runs about 0.27 mg/L lower, Old Mill (Sunken Gardens) Spring about 0.36 mg/L lower, and Upper Barton Spring about 1.21 mg/L higher. All significant at $p < 0.001$. Upper Barton Spring stands out as the clear outlier here, with a DO coefficient roughly three to four times larger in magnitude than the other two site effects. The model as a whole explains a modest but real share of the variation in DO ($R^2 = 0.20$), leaving open the question of what's driving the remaining variation as well as whether the site effect itself is genuine or partly a stand-in for something else, like survey effort or salamander detection rates. That's what the combined model below attempts to test directly.

```
# Calculating RMSE
sqrt(mean(resid(lin_model)^2))
```

```
## [1] 1.097667
```

The RMSE of about 1.10 mg/L gives a sense of typical prediction error in the model's own units. On average, predictions for DO are off by roughly 1.1 mg/L. This is a meaningful chunk of the DO range seen across these sites (mostly 0–13 mg/L, aside from the single Eliza Spring outlier). This is the in-sample error; the next step checks how well the model holds up on data it hasn't seen, using cross-validation.

```
# Number of folds
k = 5

# Setting a seed
set.seed(322)

# Randomly order rows in the dataset
shuffled_data <- model_water[sample(nrow(model_water)),]

# Create k folds in the dataset
shuffled_data <- shuffled_data |>
  mutate(folds = cut(seq(1:nrow(shuffled_data)), breaks = k, labels = FALSE))

# Initialize a vector to keep track of the performance for each k-fold
perf_k <- NULL
```

```

# Use a for-loop to get performance for each k-fold
for(i in 1:k){
  # Split data into train and test data
  train_i <- shuffled_data |> filter(folds != i) #train data = all observations except in fold i
  test_i <- shuffled_data |> filter(folds == i) # test data = observations in fold i

  # Train model on train data (all but fold i)
  train_model <- lm(DO ~ site + sampleddate,
                    data = train_i)

  # Save the RMSE for this fold
  perf_k[i] <- sqrt(mean(resid(train_model)^2))
}

# Performance for each fold
perf_k

```

```
## [1] 1.142751 1.120624 1.108006 1.021135 1.089821
```

```
# Average performance over all k folds
```

```
mean(perf_k)
```

```
## [1] 1.096467
```

The average cross-validated RMSE across the 5 folds (~1.10 mg/L) is nearly identical to the in-sample RMSE reported above. This is a good sign: it means the model isn't overfitting to the training data, and the relationship between site, sampleddate, and DO generalizes reasonably well to data the model hasn't seen. Site is a robust, if modest, predictor of DO. A pattern the combined model below now attempts to test for confounding.

```

# Getting rid of NA values - now requires DO, salamanders, and area all present
model_water <- water_cleaned |>
  filter(!is.na(DO), !is.na(salamanders), !is.na(area))

# Fit combined model
lin_model <- lm(DO ~ sampleddate + site + salamanders + area, data = model_water)
summary(lin_model)

```

```

##
## Call:
## lm(formula = DO ~ sampleddate + site + salamanders + area, data = model_water)
##
## Residuals:
##      1      2      3      4      5      6
## -3.481e-03  3.438e-03 -1.385e-18 -8.430e-20  3.590e-03 -3.547e-03
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -6.202e+01  9.159e-01  -67.71  0.00940 **
## sampleddate      3.991e-03  5.484e-05   72.77  0.00875 **
## siteUpper Barton Spring  1.281e+00  8.685e-03  147.44  0.00432 **
## salamanders    -6.601e-02  9.941e-03   -6.64  0.09516 .
## area           1.950e-02  2.022e-04   96.43  0.00660 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```
## Residual standard error: 0.007029 on 1 degrees of freedom
## Multiple R-squared:      1, Adjusted R-squared:  0.9999
## F-statistic: 2.249e+04 on 4 and 1 DF,  p-value: 0.005001
```

```
# Sample size actually used
nrow(model_water)
```

```
## [1] 6
```

This model demonstrates the approach we would use to formally test confounding. Adding salamanders and area as covariates alongside site and sampledate lets us see whether the site coefficients from the DO-only model shrink once detection rate and survey effort are accounted for. However, the output above cannot be trusted as a real result. Requiring DO, salamanders, and area to all be recorded on the same site visit leaves only 6 observations in the entire dataset, and only one site (Upper Barton Spring) even appears in the model. Eliza Spring and Old Mill drop out entirely. With 4 predictors fit to 6 points, the model has just one residual degree of freedom, which is why R-squared reaches 1 and the standard errors are implausibly small: the model is fitting noise perfectly, not capturing a real relationship.

6 Discussion

This analysis set out to answer whether dissolved oxygen (DO) varies across the four Barton Springs sites, and whether that variation is confounded by survey effort (area sampled) or salamander detection rates. The baseline model confirms the first half of this question clearly: DO differs significantly by site, even after accounting for the overall downward trend in DO over time. Upper Barton Spring stands out in particular, running roughly 1.2 mg/L higher than Barton Spring on average, a much larger effect than the differences seen at Eliza Spring or Old Mill (Sunken Gardens) Spring. This tells us site is a real, if modest, predictor of DO (R-squared = 0.20), and that this relationship holds up under cross-validation rather than being an artifact of overfitting.

The second half of the question, whether this site effect is confounded by survey effort or salamander detection, could not be answered with a formal combined model. Attempting to add area and salamanders as covariates alongside site and sampledate reduced the usable sample to just 6 site visits, since DO, area, and salamander counts are rarely all recorded on the same day at the same site. With more predictors than usable degrees of freedom, the resulting model fit the 6 points perfectly (R-squared = 1) but carries no real information about the broader dataset, and only one site even remained represented in it.

In the absence of a formal test, the descriptive patterns established earlier in this analysis offer a reasonable, if less rigorous, answer. Barton Spring is the clear outlier in survey effort, with a much wider and more variable sampled area than the other three sites (including a cluster of unusually large surveys in late 2012/early 2013). Yet Barton Spring's DO coefficient sits roughly in the middle of the four sites, not at either extreme. Therefore, a typical survey effort doesn't obviously translate into an a typical DO reading. Similarly, Eliza Spring is the clear outlier in salamander detection, accounting for the overwhelming majority of sightings and all of the highest counts in the dataset, yet its DO coefficient is only modestly below Barton Spring's and not dramatically different from Old Mill's, which had almost no salamander detections at all. Flow conditions were also examined as a third possible confound, and all four sites were sampled predominantly under normal (NORM) flow, with Barton Spring showing the least variation in flow regime of the four. Flow differences are also unlikely to explain the site-level DO pattern.

Taken together, this suggests that the site-level differences in DO are not simply a byproduct of how much area was surveyed or how often salamanders were spotted. The strongest DO effect (Upper Barton Spring) doesn't line up with either of the sites that stand out on effort or detection (Barton Spring and Eliza Spring, respectively). That said, this remains a descriptive, triangulated argument rather than a statistically confirmed one; a proper test of confounding would require substantially more consistent joint recording of DO, area, and salamander counts than this dataset currently provides, which stands as the clearest direction for future data collection or a follow-up analysis using a different modeling approach (e.g., one that doesn't require complete cases across all four variables simultaneously).

A secondary limitation worth naming: even where data existed, salamander counts are a right-skewed count variable with a large number of zeros, which fits uneasily into an ordinary linear regression framework. A count-based model (e.g., Poisson or negative binomial) would likely be a better fit for that variable in future work, though the severe missingness problem would still need to be resolved first.

7 Reflection, Acknowledgements, & References.

Huge thank you to Mr.Robert Clayton, Nick Gonzalez, and Dr. Layla Guyot who's help was integral to this work.